

Physics Informed Neural Modeling of Melt Front Dynamics in Fast Laser Melting

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Abstract—Accurate prediction of melt front propagation is crucial for understanding and controlling fast laser melting processes. This paper presents a physics informed neural network (PINN) approach for modeling the moving melt interface in a one dimensional two phase Stefan problem. The model predicts the melt front position by embedding heat conduction equations, boundary conditions, interface temperature continuity and the Stefan condition. The approach is evaluated using analytical estimates and finite difference method simulations for silver, aluminum, copper, titanium and titanium alloy. For the pure metals, the neural model closely follows the finite difference reference, with final melt front deviations from 0.11 to 3.47 %. For the titanium alloy under different laser intensities, the final deviations remain within 0.03 to 1.95 %. The results show that PINN modeling can reproduce the main melt front dynamics and provides a flexible basis for future adaptive laser control.

Index Terms—physics-informed neural networks (PINN), Stefan problem, laser melting, melt front prediction, finite difference method, Ti-6Al-4V

I. INTRODUCTION

Fast laser melting is a key physical process in laser surface treatment, laser welding, laser metal deposition, and metal additive manufacturing. During laser-material interaction, a high-energy beam rapidly heats a localized surface region and may create a moving solid-liquid interface. The depth and evolution of this melt front influence thermal history, heat-affected zone size, microstructural transformation, and final material quality. Therefore, the ability to predict the melt front position is important for process design and for future closed-loop control. The mathematical modeling of laser-induced melting is commonly based on heat conduction with phase change. Tokarev and Kaplan developed an analytical model for time-dependent pulsed laser melting by solving a one-dimensional heat conduction problem with a moving phase boundary [1]. Ngwenya and Kahlen explored analytical melt-depth prediction inside Ti-6Al-4V during laser materials processing and showed how the melt depth depends on irradiation time and laser intensity [2]. However, analytical models usually require simplifying assumptions, such as one-dimensional conduction, constant thermophysical properties, and idealized heat flux conditions. Numerical methods provide a more flexible alternative. Finite difference and finite element formulations can model transient heat transfer and phase transitions.

However, they require spatial and temporal discretization, stability control, and careful treatment of moving boundaries. Koroleva and Mazhukin explored mathematical simulation of laser-induced melting and evaporation of multilayer materials using a Stefan-type formulation with explicit tracking of phase fronts [3]. Physics-informed neural network (PINN)-based approaches have also been applied to free-boundary and Stefan problems. Wang and Perdikaris applied deep learning to free-boundary and Stefan problems by approximating both the solution and moving boundary with neural networks [4]. More recent studies have extended PINN-based approaches to two-dimensional Stefan problems [5], two-phase moving-interface problems [6], and laser-based additive manufacturing [7], [8]. The objective of this paper is to evaluate a PINN-based model for melt front propagation in fast laser melting. The process is formulated as a one-dimensional two-phase Stefan problem. The proposed neural model is compared with analytical estimates, explicit finite difference simulations, and implicit finite difference simulations for Ag, Al, Cu, Ti, and Ti-6Al-4V.

II. RELATED WORK

The Stefan problem is a classical free-boundary problem describing heat transfer with phase change. Classical formulations and theoretical foundations are discussed in standard works by Rubinstein [9] and Gupta [10]. In a Stefan problem, the temperature field and the moving interface position must be determined simultaneously. Tokarev and Kaplan presented a time-dependent analytical model for pulsed laser melting and investigated the dependence of melt depth on laser intensity [1]. Every et al. explored laser thermoelastic generation in metals above the melt threshold and considered transient thermal behavior with melting effects [11]. Ngwenya and Kahlen focused specifically on Ti-6Al-4V and developed analytical expressions for melt-depth prediction under different laser intensities [2]. Koroleva and Mazhukin considered a multifront Stefan problem for laser-induced melting and evaporation of multilayer materials [3]. Such numerical approaches can resolve the transient temperature field and moving interface, but the solution quality depends on grid resolution, time-step stability and the method used to represent latent heat. Li et al.

improved PINN performance for two-dimensional Stefan problems by combining the method with small sample learning [5]. Chang et al. proposed a physics-informed machine learning framework for two-phase moving interface Stefan problems [6]. Maidanova also investigated two-dimensional two-phase direct Stefan problems using PINNs [12]. In additive manufacturing, Zhu et al. used PINNs to predict temperature and melt pool fluid dynamics [7], while Chowdhury et al. applied PINNs to thermal field and melt pool characterization in laser metal deposition [8]. Compared with these papers, the present study focuses on a compact one-dimensional laser melting formulation and directly compares analytical, finite difference and PINN predictions for several metals and a titanium alloy.

III. MATHEMATICAL FORMULATION

The laser melting process is modeled as a one-dimensional two-phase Stefan problem. The spatial coordinate z is measured from the laser-irradiated surface into the material. The liquid region occupies $0 \leq z < X(t)$, while the solid region occupies $z > X(t)$, where $X(t)$ is the moving melt front. The temperature in the liquid phase satisfies

$$\frac{\partial T_l}{\partial t} = \alpha_l \frac{\partial^2 T_l}{\partial z^2}, \quad 0 \leq z < X(t), \quad (1)$$

and the temperature in the solid phase satisfies

$$\frac{\partial T_s}{\partial t} = \alpha_s \frac{\partial^2 T_s}{\partial z^2}, \quad z > X(t), \quad (2)$$

where T_l and T_s are the liquid and solid temperatures and α_l and α_s are the thermal diffusivities of the liquid and solid phases.

At the moving interface, the temperature is equal to the melting temperature:

$$T_l(X(t), t) = T_s(X(t), t) = T_m. \quad (3)$$

The interface velocity is governed by the Stefan condition:

$$\rho L \frac{dX}{dt} = k_s \frac{\partial T_s}{\partial z} \Big|_{X(t)^+} - k_l \frac{\partial T_l}{\partial z} \Big|_{X(t)^-}, \quad (4)$$

where ρ is density, L is latent heat of fusion, and k_s and k_l are the thermal conductivities of the solid and liquid phases.

At the laser-irradiated surface, an absorbed heat flux is imposed:

$$-k_l \frac{\partial T_l}{\partial z} \Big|_{z=0} = AI, \quad (5)$$

where A is absorptivity and I is laser intensity. Far from the interface, the solid remains close to the initial temperature:

$$T_s(z, t) \rightarrow T_0, \quad z \rightarrow \infty. \quad (6)$$

The initial state is assumed to be

$$T_l(z, 0) = T_s(z, 0) = T_0, \quad X(0) = 0. \quad (7)$$

IV. METHODOLOGY

A. Analytical Estimate

An analytical estimate is used as a classical reference for melt-depth trends. Before melting, the surface temperature can be approximated by the one-dimensional semi-infinite heat conduction solution

$$T_{\text{surf}}(t) = T_0 + \frac{2AI}{k_s} \sqrt{\frac{\alpha_s t}{\pi}}. \quad (8)$$

The approximate melting onset time is

$$t_{\text{melt}} = \frac{\pi}{4\alpha_s} \left(\frac{k_s(T_m - T_0)}{AI} \right)^2. \quad (9)$$

For $t > t_{\text{melt}}$, the analytical melt depth is estimated using an inverse complementary error function form:

$$X_a(t) = 2\sqrt{\alpha_s t} \operatorname{erfc}^{-1} \left(\frac{T_m - T_0}{T_{\text{surf}}(t) - T_0} \right). \quad (10)$$

This expression is treated as a trend-level analytical estimate rather than an exact solution for all regimes. The full two-phase PINN and FDM formulations include additional interface and latent heat constraints.

B. Finite Difference Reference

The finite difference method (FDM) is used as the numerical reference. The explicit FDM solution is used as the main reference curve, while the implicit FDM solution is included to check numerical consistency. The computational domain is discretized in space and time, and the phase change is represented through a latent-heat formulation. The explicit scheme uses a stability-restricted time step, while the implicit scheme provides a more stable comparison under refined temporal settings. The numerical solution tracks the moving melt front $X_{\text{FDM}}(t)$ and provides the reference trajectory for evaluating PINN predictions. The relative final deviation is computed as

$$\Delta_{\text{final}} = \frac{|X_{\text{PINN}}(t_f) - X_{\text{FDM}}(t_f)|}{X_{\text{FDM}}(t_f)} \times 100\%. \quad (11)$$

The relative L_2 error is computed over the full time interval:

$$E_{L_2} = \frac{\|X_{\text{PINN}} - X_{\text{FDM}}\|_2}{\|X_{\text{FDM}}\|_2} \times 100\%. \quad (12)$$

C. Physics Informed Neural Network

The PINN approximates the liquid temperature, solid temperature and moving interface as

$$T_l(z, t) \approx T_l^\theta(z, t), \quad T_s(z, t) \approx T_s^\theta(z, t), \quad X(t) \approx X^\beta(t), \quad (13)$$

where θ and β denote trainable neural-network parameters. Automatic differentiation is used to compute spatial and temporal derivatives.

The total loss function is constructed as

$$\mathcal{L} = \lambda_l \mathcal{L}_{r,l} + \lambda_s \mathcal{L}_{r,s} + \lambda_q \mathcal{L}_q + \lambda_f \mathcal{L}_\infty + \lambda_T \mathcal{L}_{T_m} + \lambda_X \mathcal{L}_S + \lambda_0 \mathcal{L}_0, \quad (14)$$

where $\mathcal{L}_{r,l}$ and $\mathcal{L}_{r,s}$ enforce the liquid and solid heat equations, $\mathcal{L}_q$ enforces the surface heat flux, $\mathcal{L}_\infty$ enforces the far-field condition, $\mathcal{L}_{T_m}$ enforces interface temperature continuity, $\mathcal{L}_S$ enforces the Stefan condition and $\mathcal{L}_0$ enforces the initial condition.

The liquid and solid PDE residual losses are

$$\mathcal{L}_{r,l} = \frac{1}{N_l} \sum_{i=1}^{N_l} \left| \frac{\partial T_l^\theta}{\partial t} - \alpha_l \frac{\partial^2 T_l^\theta}{\partial z^2} \right|^2, \quad (15)$$

$$\mathcal{L}_{r,s} = \frac{1}{N_s} \sum_{i=1}^{N_s} \left| \frac{\partial T_s^\theta}{\partial t} - \alpha_s \frac{\partial^2 T_s^\theta}{\partial z^2} \right|^2. \quad (16)$$

Let $X_i = X^\beta(t_i)$. The interface temperature loss is defined as

$$\mathcal{L}_{T_m} = \frac{1}{N_X} \sum_{i=1}^{N_X} \left[\left| T_l^\theta(X_i, t_i) - T_m \right|^2 + \left| T_s^\theta(X_i, t_i) - T_m \right|^2 \right]. \quad (17)$$

The Stefan loss is

$$\mathcal{L}_S = \frac{1}{N_X} \sum_{i=1}^{N_X} \left| \rho L \frac{dX^\beta}{dt}(t_i) - k_s \frac{\partial T_s^\theta}{\partial z}(X_i, t_i) + k_l \frac{\partial T_l^\theta}{\partial z}(X_i, t_i) \right|^2. \quad (18)$$

This formulation allows the network to learn a continuous representation of the temperature fields and melt-front motion while remaining constrained by the physical model.

V. EXPERIMENTAL SETUP

The experiments are divided into two groups. The first group considers Ag, Al, Cu and Ti under a laser intensity of 10^9 W/m² over a final time of 10 s. The second group considers Ti-6Al-4V under laser intensities of 5, 50, 500 and 5000 kW/cm² over a final time of 7 microseconds. The conversion $1 \text{ kW/cm}^2 = 10^7 \text{ W/m}^2$ is used where necessary.

Table I summarizes the main material parameters used in the simulations.

The pure-metal simulations compare the analytical estimate, explicit FDM, implicit FDM and PINN. The Ti-6Al-4V simulations compare the FDM solutions and PINN with the analytical reference based on the melt-depth formulation used in laser processing studies of this alloy.

VI. RESULTS AND DISCUSSION

A. Pure Metals

Fig. 1 shows the predicted melt-front trajectories for Ag, Al, Cu and Ti at $I = 10^9$ W/m². The analytical estimate predicts a larger melt depth than the numerical and neural solutions for Ag, Al and Cu. This difference is expected, since the analytical

model is based on simplifying assumptions, while the FDM and PINN models directly enforce the two-phase formulation and the latent heat condition.

The PINN predictions closely follow the explicit FDM reference for all four materials. Table II summarizes the final deviation and relative L_2 error. The final deviation remains below 3.5% in all pure-metal cases. Ag shows the lowest final deviation, while Ti shows the highest final deviation. However, even in the Ti case, the final discrepancy remains small relative to the total melt depth.

The error curves in Fig. 1 indicate that the largest deviations generally occur during the early or intermediate stage of melting. This is a typical difficulty in moving-boundary problems because the interface begins from a near-zero position and its velocity changes rapidly. After this transient stage, the PINN predictions stabilize and remain close to the FDM solution.

B. Ti-6Al-4V Alloy

Ti-6Al-4V is evaluated separately because it is widely used in laser-based manufacturing and biomedical or aerospace applications. Fig. 2 shows the melt-front prediction for laser intensities of 5, 50, 500, and 5000 kW/cm². The final time is 7 microseconds.

The PINN follows the explicit FDM reference closely across all tested intensities. The analytical reference predicts a larger melt depth, especially at lower and medium intensities. The PINN and FDM curves remain closer to each other because both include the same two-phase heat conduction structure and interface energy balance.

Table III summarizes the Ti-6Al-4V errors. The final deviation ranges from 0.0% to 1.9%, and the relative L_2 error ranges from 0.6% to 2.9%. These values indicate that the PINN is able to approximate the moving melt front under both lower and higher intensity regimes.

At the beginning of the process, the error curves show short transient deviations. These deviations can be explained by the fast initial acceleration of the melt front and the difficulty of enforcing all interface conditions when the melt depth is very small. Nevertheless, the deviations decrease as the interface grows, and the final agreement with the FDM reference is strong.

C. Discussion

The results show that PINNs can serve as a useful alternative to classical numerical modeling for one-dimensional laser melting problems. The main advantage is that the neural network provides a continuous differentiable representation of temperature fields and interface position. This is useful for future inverse modeling and control tasks because derivatives and sensitivities can be obtained through automatic differentiation. The comparison also shows that analytical estimates are valuable for interpreting general trends but may overestimate melt depth compared with FDM and PINN predictions. This behavior is visible in both the pure-metal and Ti-6Al-4V experiments. Therefore, analytical models should be used as simplified references, while FDM and PINN models are

TABLE I
MATERIAL PARAMETERS USED IN THE SIMULATIONS

Material	ρ kg/m ³	k_s W/(m K)	k_l W/(m K)	α_s m ² /s	α_l m ² /s	T_m K	L J/kg	A
Ag	10500	429.0	361.0	1.7375×10^{-4}	1.3309×10^{-4}	1234	1.112×10^5	0.020/0.043
Al	2700	238.0	100.0	9.6130×10^{-5}	3.8807×10^{-5}	933	3.880×10^5	0.0588/0.064
Cu	8960	401.0	342.0	1.1611×10^{-4}	8.9060×10^{-5}	1358	2.047×10^5	0.020/0.058
Ti	4500	21.6	20.28	9.0900×10^{-6}	7.0550×10^{-6}	1940	3.650×10^5	0.257/0.433
Ti-6Al-4V	4510	20.0	29.0	5.8000×10^{-6}	5.9500×10^{-6}	1928	2.900×10^5	0.433

Ag / Al / Cu / Ti | $I = 10^9$ W/m²
FDM explicit (reference) · FDM implicit · PINN · Analytical

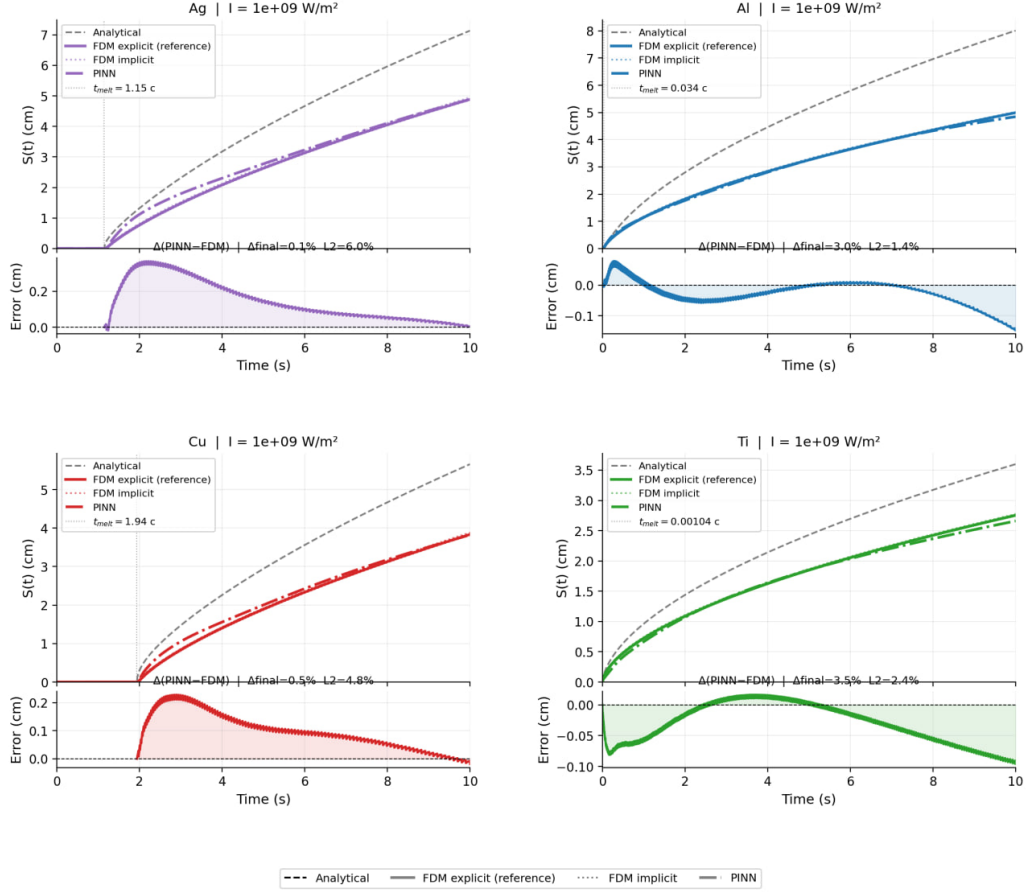


Fig. 1. Comparison of analytical, explicit FDM, implicit FDM and PINN melt-front predictions for Ag, Al, Cu, and Ti at $I = 10^9$ W/m².

TABLE II
PINN DEVIATION FROM EXPLICIT FDM REFERENCE FOR PURE METALS

Material	FDM cm	PINN cm	Δ_{final} %	L_2 %
Ag	4.8842	4.8895	0.11	5.99
Al	4.9925	4.8443	2.97	1.44
Cu	3.8430	3.8244	0.48	4.81
Ti	2.7563	2.6608	3.47	2.41

TABLE III
PINN DEVIATION FROM EXPLICIT FDM REFERENCE FOR Ti-6Al-4V

Intensity	FDM μ m	PINN μ m	Δ_{final} %	L_2 %
5 kW/cm ²	6.767	6.899	1.95	2.04
50 kW/cm ²	16.525	16.551	0.15	0.65
500 kW/cm ²	23.801	23.829	0.12	1.51
5000 kW/cm ²	29.844	29.854	0.03	2.88

more suitable for detailed comparison under the two-phase

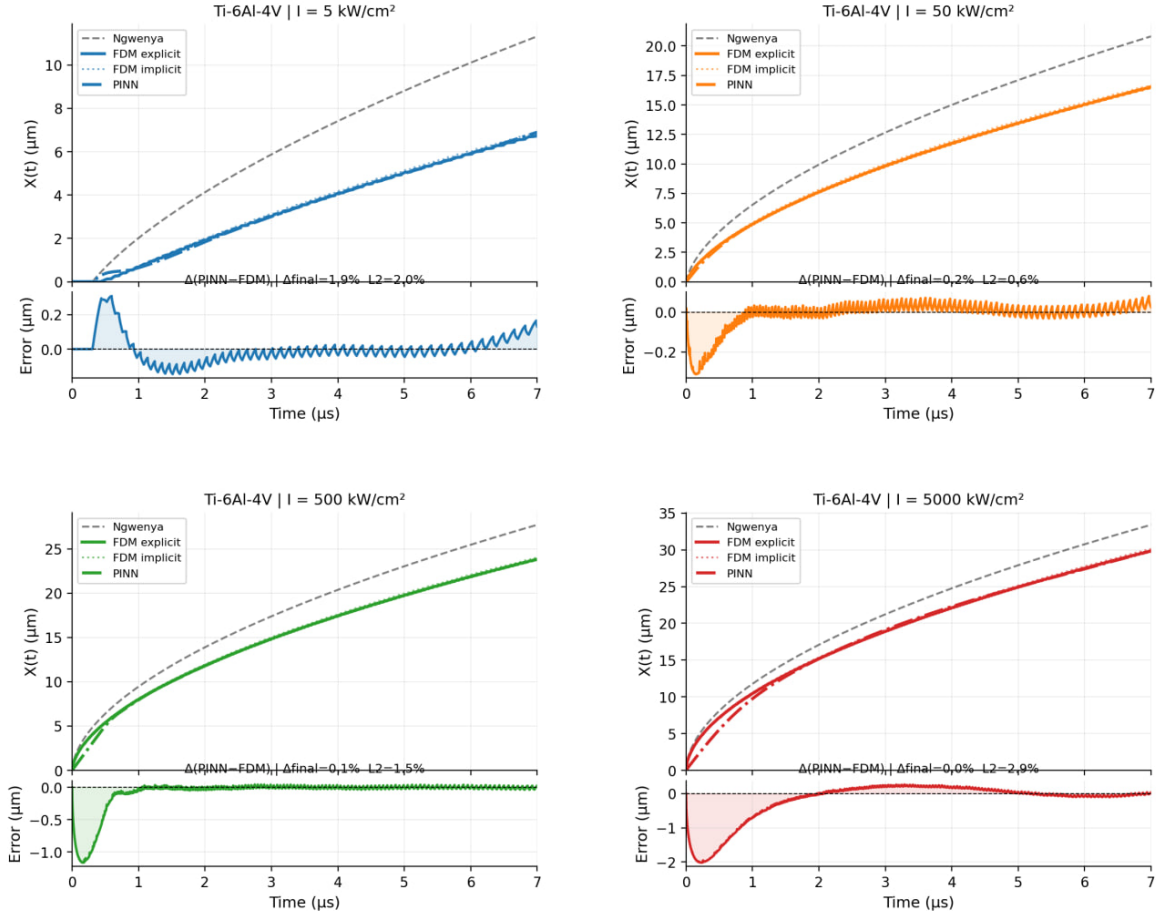


Fig. 2. Comparison of analytical, explicit FDM, implicit FDM, and PINN melt-front predictions for Ti-6Al-4V under different laser intensities.

formulation. The present model has several limitations. First, the formulation is one-dimensional and assumes conduction-dominated heat transfer. Second, thermophysical properties are treated as constant within each phase. Third, hydrodynamic melt flow, evaporation, recoil pressure, surface tension and Marangoni effects are not included.

VII. CONCLUSION

This paper presented a physics informed neural network approach for modeling melt-front propagation in fast laser melting. The process was formulated as a one-dimensional two-phase Stefan problem, with heat conduction in liquid and solid phases, temperature continuity at the moving interface, laser heat flux at the surface and the Stefan condition governing interface motion. The model was compared with analytical estimates, explicit FDM and implicit FDM for Ag, Al, Cu, Ti and Ti-6Al-4V. For the pure metals, the final deviation between PINN and explicit FDM ranged from 0.11% to 3.47%. For Ti-6Al-4V under laser intensities from 5 to 5000 kW/cm², the final deviation ranged from 0.03% to 1.95%. These results demonstrate that the PINN can reproduce melt-front dynamics

with good agreement relative to the finite difference reference. Future work will extend the model toward reinforcement-learning-based adaptive control. In such a framework, the laser intensity can be adjusted dynamically to guide the melt front toward a desired trajectory while maintaining physical constraints.

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